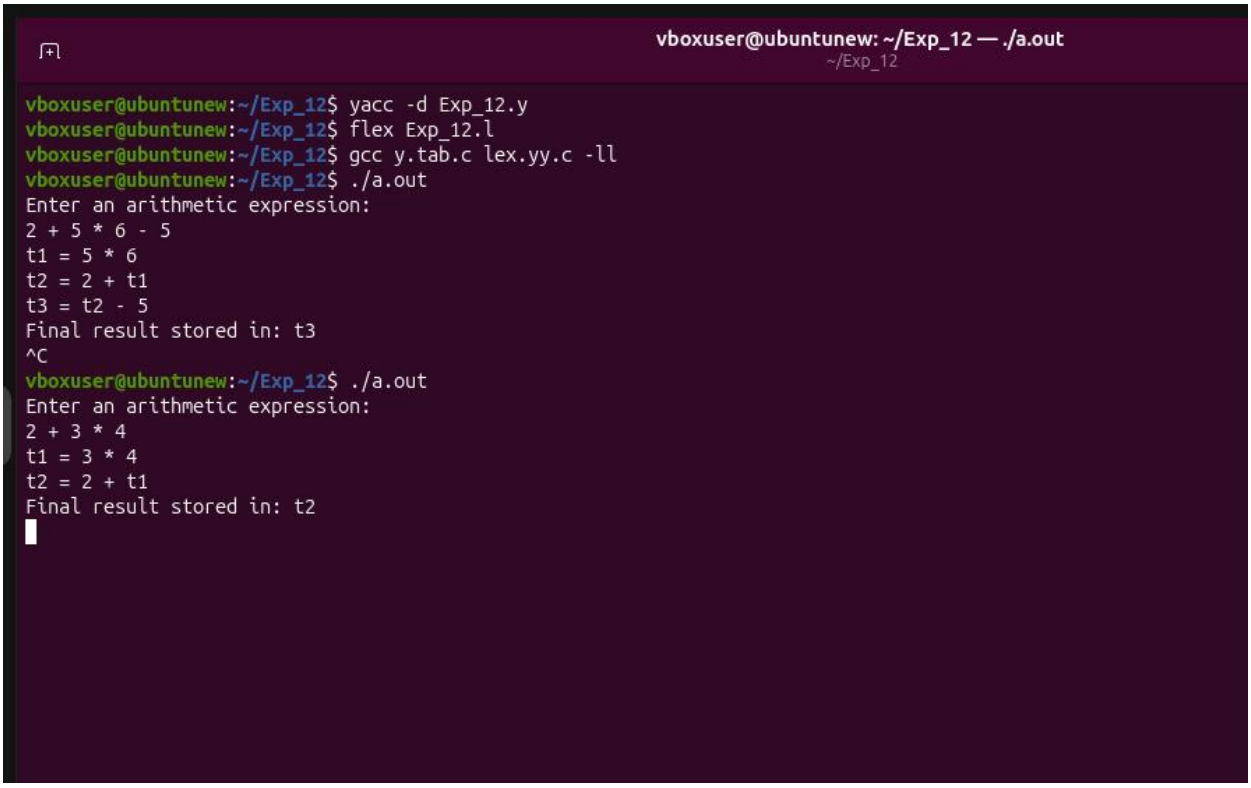
Name: Shreya Bawankar

Prn: 24070521120 Sec: B Batch: B2



Name: Shreya Bawankar

Prn: 24070521120 Sec: B Batch: B2



Name: Shreya Bawankar

Prn: 24070521120 Sec: B Batch: B2

Open

Exp_12.y
~/

Exp_12.l | Exp_12.y

```
%{
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int temp = 1;
int yylex(void);
int yyerror(char *s);
%}

%union {
    char str[20];
}

%token <str> NUMBER ID
%type <str> expr

%left '+' '-'
%left '*' '/'

%%

input:
    expr '\n'
    {
        printf("Final result stored in: %s\n", $1);
    }
    ;

expr:
      expr '+' expr
      {
          sprintf($$, "t%d", temp++);
          printf("%s = %s + %s\n", $$, $1, $3);
      }
```

Name: Shreya Bawankar

Prn: 24070521120 Sec: B Batch: B2

Open

Exp_12.y
~/

Exp_12.l | Exp_12.y

```
    expr '+' expr
    {
        sprintf($$, "t%d", temp++);
        printf("%s = %s + %s\n", $$, $1, $3);
    }
  | expr '-' expr
    {
        sprintf($$, "t%d", temp++);
        printf("%s = %s - %s\n", $$, $1, $3);
    }
  | expr '*' expr
    {
        sprintf($$, "t%d", temp++);
        printf("%s = %s * %s\n", $$, $1, $3);
    }
  | expr '/' expr
    {
        sprintf($$, "t%d", temp++);
        printf("%s = %s / %s\n", $$, $1, $3);
    }
  | '(' expr ')'
    {
        strcpy($$, $2);
    }
  | ID
    {
        strcpy($$, $1);
    }
  | NUMBER
    {
        strcpy($$, $1);
    }
  ;

%%
```

```c
int main()
{
    printf("Enter an arithmetic expression:\n");
    yyparse();
    return 0;
}

int yyerror(char *s)
{
    printf("Invalid expression\n");
    return 0;
}
```